Bug Hunters

KRITICAL HACKS 2026

IGDTUW Delhi · AI/ML Track (Open Innovation) · Student category

Project Title : **TakeSure — Confidence before taking something.** Ayurveda × Allopathy herb–drug interaction checker: photo of prescription + photo of product → sourced, plain-language safety verdict + Ministry of Ayush ADR report.

Team Name : ***Bug Hunters***

Team Members :

Dhriti Garg — Team lead · backend, rule engine, medical research
Shiwani Goyal — Frontend · web app, Hindi/English UI, result card, demo video & pitch
Khushi Gupta — Knowledge base & evidence grading, OCR, PDF/ADR export, TinyFish curation

Track Chosen : **AI/ML Track — Open Innovation · Healthcare**

TinyFish : **Mandatory integration implemented —** Fetch API verifies every citation · Web Agent drafts interaction rows from monographs · Search + Agent map Indian products to herbs. Human-reviewed, off the verdict path.

Status : Working prototype: 25 API routes · 49 cited interactions · 78 sources · 58 automated tests · offline-capable.

Aligned with the Ministry of Ayush Pharmacovigilance Programme for ASU&H drugs · decision support, never diagnosis

PROBLEM STATEMENT & TARGET AUDIENCE

PROBLEM WE'RE SOLVING

1. **India runs two medical systems side by side — and nobody checks the combination.** A thyroid patient on Thyronorm adds ashwagandha for stress; a transplant patient on tacrolimus drinks giloy kadha; a cardiac patient on warfarin takes guggul for cholesterol. Ashwagandha can raise thyroid hormone, giloy can oppose immunosuppression, guggul can increase bleeding.

2. **The allopathic doctor rarely asks, the patient rarely tells, the chemist has no tool,** and the product label says nothing about allopathic medicines.

3. **Existing checkers (Medscape, Drugs.com, 1mg) only know allopathic × allopathic pairs;** Western herb databases are English-only, paid and do not know Indian brands like Himalaya Ashvagandha or Dabur Giloy Ghanvati.

37%	of hospital patients used allopathic + herbal medicines together
65%	of herbal users lacked knowledge of dose, side effects, interactions
19%	of Indian patients disclose herbal use to their doctor
1,473	ASU&H adverse-event reports nationally since 2018

Impacted: patients on thyroid, diabetes, BP/heart, blood-thinner, transplant or epilepsy medicines and their families; doctors who never hear about the co-use; the Ministry's pharmacovigilance network that never receives the reports.

TARGET USERS

B2C — Patients & caregivers

Adults 30–70 on long-term medicines who also take Ayurvedic products; family members managing an elder's medicines; Hindi or English, phone-first.

B2B — Clinicians & pharmacies

MBBS/MD doctors, BAMS vaidyas, hospital pharmacies, retail chemists, telemedicine and hospital-HIS platforms (API).

B2G — Regulators

Ministry of Ayush / NPvCC at AIIA and the 99 Peripheral Pharmacovigilance Centres: structured ICSRs and co-use analytics.

TAKESURE — OUR UNIQUE SOLUTION

Key Features

1. **Scan —** Photograph the prescription and the Ayurvedic product label (or type names). OCR reads them; every name is shown for confirmation before anything is checked.

2. **Match —** Indian brand → generic (Thyronorm → levothyroxine), product → herbs (Chyawanprash → amla, ashwagandha…) via a curated bridge dictionary + fuzzy matching.

3. **Decide —** Deterministic rule engine over a cited knowledge base → GREEN / AMBER / RED / NOT IN DATABASE, with timing gaps (e.g. keep 2–3 h between Triphala and oral medicines).

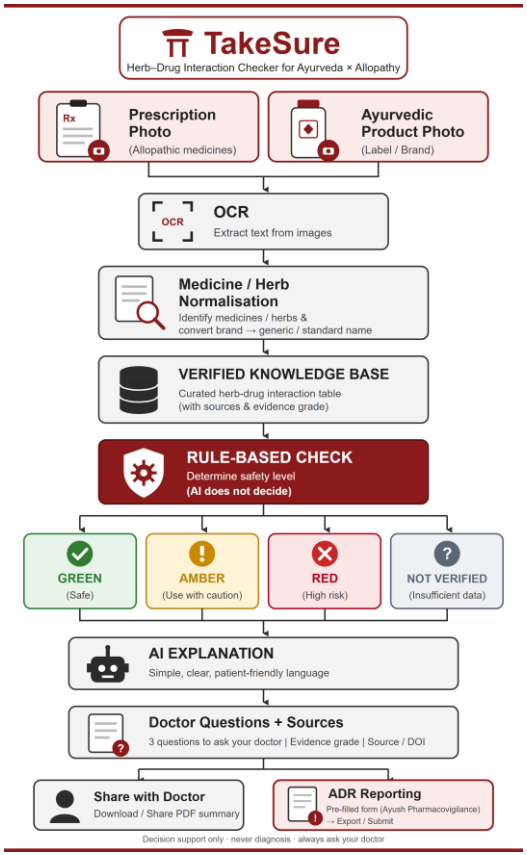
4. **Explain —** AI rewrites only the matched evidence into simple Hindi/English + 3 questions to ask the doctor. It cannot add, remove or re-grade an interaction.

5. **Report —** One-page PDF with sources for the doctor; one-tap suspected-ADR report in the Ministry of Ayush format for the Ayush Suraksha portal.

Novelty

- **"Verdict from published data; AI only explains."** The risk level never comes from a language model — it comes from a versioned, cited knowledge-base row. Turn the AI off and the verdict is identical.
- **The missing bridge table:** Indian brand names × Ayurvedic herb synonyms (Sanskrit/Hindi/Latin) × graded published evidence — open, auditable, co-curated with Ayurveda colleges.

Project workflow



ROLE OF AI / ML

AI everywhere it is safe. Rules in the one place it is dangerous.

Computer vision	**OCR**	Tesseract 5.4 (pre-trained LSTM) reads prescription and label photos after Pillow pre-processing (grayscale, upscale, autocontrast); English + Hindi models. Roadmap: fine-tune on Indian labels and handwriting.
NLP	**Name matching**	Entity extraction from messy prescription text; RapidFuzz token matching over a bridge dictionary of Indian brands, generics and Sanskrit/Hindi/Latin synonyms. ≥85 accept, 70–85 ask the user, <70 unrecognised.
LLM	**Explainer**	Groq-hosted open-weight model (gpt-oss-120b), temperature 0, receives only the matched rows, must return a strict JSON schema; directive language rejected in English and Hindi; knowledge-base fallback. Output: plain-language summary + 3 doctor questions.
Agentic AI	**TinyFish research assistant**	Web agents fetch citations, read monographs and product pages, and draft knowledge-base rows with sources attached. Human-reviewed. This is how the dataset scales.
Rules	**Verdict engine**	Deterministic rules R1–R5 over cited rows: severity + evidence grade + drug class → GREEN / AMBER / RED / NOT IN DB. Reproducible, auditable, works with every model switched off.

Why no trained classifier for the verdict: no labelled dataset exists (≈1,473 national reports in 7 years), a classifier cannot explain itself to a doctor or regulator, and medical decision support must be reproducible. Future ML once data exists: OCR fine-tuning on Indian labels, a named-entity model for prescriptions, and a model that ranks under-reported pairs for curation priority.

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TECH STACK + ARCHITECTURE

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TECH STACK

FE — Frontend
Next.js (React) + TypeScript, Tailwind CSS, Framer Motion · installable PWA · Hindi/English UI

BE — Backend
FastAPI (Python), Pydantic v2 validation, SQLAlchemy · REST API with typed errors

DB — Database
PostgreSQL (Supabase) in production; SQLite for offline/demo · tables: herbs, drugs, products, interactions, sources

OCR — OCR
Tesseract (eng + hin) on the server; Google Vision / Groq vision model as optional cloud reader

FZ — Matching
RapidFuzz token-set matching; ≥85 accept, 70–85 ask the user, <70 not recognised

AI — AI explainer
Groq API (Llama / GPT-OSS) with JSON-schema output, temperature 0, facts-only prompt, fallback text

PDF — PDF & hosting
ReportLab for doctor summary + ADR form · Vercel (web) + Render/Railway (API)

TF — TinyFish curation
Fetch verifies citations · Web Agent drafts rows from monographs · Search+Agent map products → human-approved drafts

ARCHITECTURE

1 Inputs
Prescription + product label photos, or typed names

2 OCR
Tesseract eng+hin / vision model · file deleted after read

3 Matcher
RapidFuzz · brand→generic · product→herbs · user confirms

4 Rule engine (decides)
Rules R1–R5: severity + grade → GREEN / AMBER / RED / NOT IN DB

5 Explainer (AI)
Groq LLM · matched rows only · JSON schema · cannot change verdict

6 Outputs
Result card (Hindi/English) · doctor PDF · ADR report

7 Anonymised log
ids + verdict + state → dashboard for Ministry / NPvCC

DB Knowledge base
PostgreSQL / SQLite · herb × drug rows: severity, grade A/B/C, timing, DOI/PMID

FLOW

1 Verdict path
Photos → OCR text → candidate names → user confirms → rule engine looks up every herb × medicine pair → worst row wins (RED > AMBER > GREEN; unknown is never GREEN).

2 AI guardrail
Only matched rows go to the LLM, which returns {summary_hi, summary_en, timing_gap, doctor_questions[3]}; invalid or directive output is rejected and the knowledge-base text is shown instead.

3 Privacy
Uploads are deleted after extraction; the log stores ids and verdicts only — the basis of the anonymised dashboard for the Ministry of Ayush.

4 Curation loop (TinyFish)
Offline from verdicts: Fetch verifies citations, the Web Agent drafts rows from monographs, Search+Agent map products. Every draft keeps its source URL and run id and waits for a reviewer.

TINYFISH INTEGRATION

TinyFish is TakeSure's research assistant. It opens the references, reads monographs and product pages, and drafts knowledge-base rows with the source attached. Reviewers approve every row. The verdict path never touches the live web — that is a safety rule, not a limitation.

Fetch API — **Verify citations**

Three stages: Fetch the resolved page; Fetch the Crossref record for bot-blocked DOIs; Search the title for sources without a locator. ≥60% of distinctive title terms matched → verified; otherwise a "mismatch" draft with the best candidate.

Live run 19 Oct: 55 of 78 citations verified automatically; 23 listed for review (3 broken links found). Free tier.

Fetch + Web Agent — **Discover interaction rows**

Fetch the MSKCC About Herbs page (free, 2 s) and extract every clinical bullet naming a drug, class or interaction verb from our drug dictionary; if the static page yields nothing, POST /v1/automation/run-async with a JSON output_schema and poll the run.

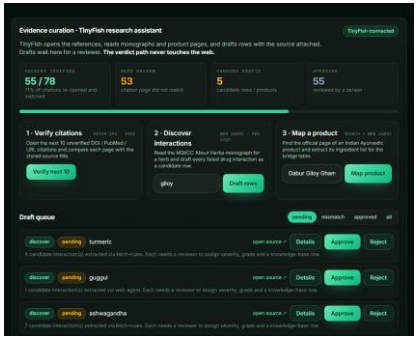
Live: 7 candidates for ashwagandha, 4 for turmeric in seconds (Fetch + rules); Web Agent fallback for guggul.

Search + Agent — **Map Indian products**

GET api.search.tinyfish.ai for the product page, then an Agent run extracts the ingredient list with Latin/Sanskrit names; matched to known herbs.

Live: Dabur Giloy Ghanvati mapped to giloy (1 g/tablet) in one Agent run (28 steps).

Evidence-curation panel on the dashboard



Live: 55/78 sources verified, 5 pending drafts, queue with approve / reject.

IMPLEMENTATION STATUS · DEMONSTRABLE

- **Backend:** app/services/tinyfish_service.py (Fetch, Search, Agent clients) · curation_drafts table · 7 routes under /api/curation · 7 offline tests with a mocked TinyFish transport (58 tests total).
- **Frontend:** "Evidence curation" panel on the dashboard with Verify / Discover / Map actions and an approve-reject draft queue; health endpoint reports tinyfish_configured.
- **Guardrail:** TINYFISH_API_KEY lives only on the server; drafts never change a verdict until a reviewer approves. Live demo: Verify next 10 → the sources-verified counter rises.

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FEASIBILITY AND SHOWSTOPPERS

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TECHNICAL

- Proven, off-the-shelf stack
- OCR: Google Vision / Paddle OCR
- RapidFuzz for name matching
- SQL for KB lookup
- LLM only explains — rules decide risk
- No custom ML training

DATA & EVIDENCE

- Public / citable sources: Ayurvedic Pharmacopoeia, NIH LiverTox, MSKCC, PubMed, WHO monographs
- Every entry stores evidence, severity & source
- Evidence graded A / B / C

RESOURCES

- Team: Backend · Mobile/Frontend · Data curation
- Local/offline components + free/low-cost infrastructure
- Swappable LLM provider

MVP SCOPE

- ~40 herb–drug pairs · 2 photo inputs · manual text-entry fallback · risk / safety card · ADR reporting form
- Not in MVP: diagnosis · dosing · herb–herb interactions · custom ML training

REGULATORY / OPERATIONAL SAFETY

- ADR workflow aligned with Ministry/ICSR fields
- Links to Ayush Suraksha instead of a parallel reporting system
- Photos not retained beyond the session unless the user opts in
- Decision support — not diagnosis or prescription

RISK / BLOCKER	MITIGATION
OCR misread Handwritten prescriptions · poor-quality or unusual printed labels	Extracted names shown for user confirmation · manual text entry always available · fuzzy-matching thresholds separate confident from uncertain matches
Unknown herb / drug Herb or drug may not be in the KB · coverage gaps	NEVER guess · neutral “NOT IN DATABASE” result · unknown items logged for future curation · MVP deliberately scoped
Weak / conflicting evidence Limited clinical evidence · sources may disagree	A/B/C grade on every result · source / reference visible · conflicting sources kept at lower confidence, not hidden · KB entries reviewed before publication
LLM hallucination Unsupported medical claims	LLM receives ONLY matched KB info · never determines severity or verdict · schema-validated output · raw KB text if validation fails · safety decision works with LLM off
Medical liability / safety scope Users may read it as a prescription or diagnosis tool	Positioned as DECISION SUPPORT · never says stop / start · no diagnosis or dosing · evidence + sources shown · questions for the doctor · report-incorrect-info mechanism
API / connectivity failure Internet or external API / provider failure	Local / offline fallback · SQLite KB · PaddleOCR runs locally · LLM provider swappable · DB lookup independent of LLM · cached outputs for demo continuity

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USP (UNIQUE SELLING PROPOSITION) & BUSINESS MODEL

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USP (UNIQUE SELLING PROPOSITION)

1. **India's first safety layer between the two medical systems people already mix** — nothing public today checks an Ayurvedic product against an allopathic prescription.
2. **Verdict from published data; AI only explains** — auditable rules over cited rows; the opposite of a "ChatGPT wrapper".
3. **Built for India:** Indian brand names, Sanskrit/Hindi synonyms, Hindi explanations, Ministry of Ayush ADR format, Ayush Suraksha portal link.
4. **Honest by design:** evidence grades, timing gaps instead of alarms, "not in database" instead of guesses.

BUSINESS MODEL

B2C	**Free for citizens** — Web/PWA checker in Hindi & English; grows the knowledge base and the ADR signal.
B2B	**API & counter tool** — Per-query / subscription API for pharmacy chains, hospital HIS, telemedicine apps; QR on the chemist's bill.
B2G / CSR	**Programmes & grants** — Co-use analytics and structured ICSRs for the Ministry of Ayush pharmacovigilance network; CSR funding from Ayurvedic brands that want a trust layer.
PLATFORM	**Ayush Grid / ABDM integration** — verdict attached to the ABHA health record; open KB co-curated with Ayurveda colleges.

WHO BENEFITS · ONE KNOWLEDGE BASE, FOUR USERS

PATIENTS & FAMILIES	DOCTORS & VAIDYAS	PHARMACISTS & CHEMISTS	MINISTRY OF AYUSH / NPvCC
60-second verdict in Hindi or English, timing gaps, and three questions to carry to the clinic	Patients finally disclose what they take; a one-page summary with sources bridges both vocabularies	Counter-side cross-check for the only shop that sells both systems' products; QR on the bill	Structured ICSRs in the official format, anonymised co-use analytics by state, early signal detection

IMPACT · WHY IT MATTERS

1,473
ASU&H adverse-event reports nationally since 2018 — TakeSure turns every reaction into a structured ICSR

75%
of herbal users with a side effect never told their doctor — TakeSure hands them the questions to ask

0
public tools today check an Ayurvedic product against an allopathic prescription — TakeSure is the first

FUTURE SCOPE

ROADMAP

- **Knowledge base:** 200+ pairs; second-reviewer rule on every citation; faculty review with an Ayurveda college or pharmacology department before public release.
- **ML once data exists:** OCR fine-tuned on Indian labels and handwriting; a named-entity model for prescriptions; ranking of under-reported pairs for curation priority.
- **TinyFish:** scheduled monthly re-verification of every citation; agent monitoring of LiverTox and Ayush Suraksha for new signals; product-catalogue sweeps of Himalaya, Dabur, Patanjali, Baidyanath.
- **Integration:** Ayush Grid / ABDM so the verdict and ADR sit on the ABHA record; pharmacy-chain API; chemist QR pilot in one city.
- **Out of scope by design:** diagnosis, dosing, herb–herb interactions, any trained model deciding a verdict.

24-HOUR BUILD PLAN · 31 OCT – 1 NOV

0–3 h	Set up at venue; load KB; core loop photo → OCR → confirm → verdict green.
3–9 h	TinyFish verifier over all 78 citations; reviewer pass on mismatches.
9–15 h	Agent discovery for 16 herbs → draft rows → review → 100+ rows.
15–19 h	Product mapping for top 30 Indian products; Hindi in every PDF.
19–22 h	Public deploy (Vercel + Render + Supabase); chemist QR flow.
22–24 h	Freeze, run tests, rehearse the demo on the deployed link.

RESEARCH & REFERENCES

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1. **Sharma 2018: ashwagandha & thyroid trial**
RCT, 50 patients, 600 mg/day, 8 weeks
pubmed.ncbi.nlm.nih.gov/28829155

2. **Gohil & Patel 2007: herb–drug case reports**
133 cases, only 6% well documented (Indian J Pharmacol)
doi.org/10.4103/0253-7613.33432

3. **Ghosh et al. 2009: herb–drug interaction**
Fallout of poor doctor–patient communication (JCDR)
jcdr.net/articles/PDF/576/501.pdf

4. **Interactions between substances in Ayurveda**
Food–drug & Ayurveda–modern drug review, 2013
doi.org/10.1515/dmdi-2013-0014

5. **NSSO Survey on Ayush 2022-23 (MoSPI press note)**
Ayurveda is the most used Ayush system
mospi.gov.in/…/PressNote_Ayush_13062024N.pdf

6. **Ayush Suraksha portal (Ministry of Ayush)**
Suspected ADR reporting for ASU&H drugs
suraksha.ayush.gov.in

7. **ASU&H pharmacovigilance (ITRA, Jamnagar)**
PPvC / IPvC reporting network
itra.ac.in/pharmacovigilance

8. **Open-source tools we build on**
OCR (Tesseract) and name matching (RapidFuzz)
github.com/tesseract-ocr · github.com/rapidfuzz

9. **TinyFish documentation**
Fetch, Search and Web Agent APIs used for evidence curation
docs.tinyfish.ai

COMPARISON WITH EXISTING TOOLS

FEATURE	MEDSCAPE · DRUGS.COM · 1MG	MSKCC · NATURAL MEDICINES	TAKESURE (OUR APP)	WHY TAKESURE?
Ayurveda × allopathy check	✗ Allopathic × allopathic pairs only ★☆☆☆☆ 1 / 5	~ Single-herb monographs only ★★☆☆☆ 2 / 5	✓ Prescription × Ayurvedic product, every herb–drug pair checked ★★★★★ 5 / 5	**★ Only tool that checks the two systems people already mix.**
Indian brands & Hindi	~ Partly covered ★★★☆☆ 3 / 5	✗ Western herbs, English only ★☆☆☆☆ 1 / 5	✓ Indian brands + Sanskrit / Hindi synonyms, Hindi UI ★★★★★ 5 / 5	**★ Matches Himalaya, Dabur and the names patients actually use.**
Sources on every verdict	✓ Sources shown ★★★★☆ 4 / 5	✓ Sources shown ★★★★☆ 4 / 5	✓ Sources + grade A/B/C with DOI/PMID; marked “needs verification” until re-checked ★★★★☆ 4 / 5	**★ Weak evidence is shown as weak, never hidden — and we say so when a citation is not yet re-verified.**
ADR report to Ministry	✗ No reporting ★☆☆☆☆ 1 / 5	✗ No reporting ★☆☆☆☆ 1 / 5	✓ One-tap report in Ministry of Ayush format ★★★★★ 5 / 5	**★ Feeds the Ayush pharmacovigilance network.**
OVERALL RATING	**Medscape 2.3 / 5** ★★☆☆☆	**MSKCC 2.0 / 5** ★★☆☆☆	**TakeSure 4.8 / 5** ★★★★★	**TakeSure scores 2.5 points higher than the best alternative.**

Ratings out of 5 are the team's assessment on the four features compared (sourced = 4, +1 with evidence grade · partly = 3 · single herb = 2 · none = 1). TakeSure takes 4/5 on Sources deliberately: every citation stays marked “needs verification” until a second reviewer re-opens the DOI/PMID.